Prob. 3. (Z&T, Ch. 5, 7.1.)

Show that $u_{\xi\xi} - u_{\eta\eta} + \underbrace{(\ldots)}_{\text{lower derivative terms}} = 0$ can be derived from $u_{\xi\eta} + (\ldots) = 0$ using a $45°$ rotation in the axis (is the transformation OK? Check Jacobian). Let's begin with the xy coor-s ($u_{xy} = 0$), and go to the $\xi\eta$ coor-s ($u_{\xi\xi} - u_{\eta\eta} + \ldots = 0$). Do not use any formulas (7.7 – 7.9).

Sol-n.

Let

(1) $$u_{xy} = 0.$$

(To come to the exactly required form $u_{\xi\xi} - u_{\eta\eta} + \ldots = 0$ it turns out that one should use "<u>clockwise</u>" rotation.)

Consider

(2) $$J = \begin{vmatrix} \cos(\theta) & \sin(\theta) \\ -\sin(\theta) & \cos(\theta) \end{vmatrix}\Bigg|_{\theta = 45°} = \begin{vmatrix} \cos(45°) & \sin(45°) \\ -\sin(45°) & \cos(45°) \end{vmatrix} = \begin{vmatrix} \sqrt{2}/2 & \sqrt{2}/2 \\ -\sqrt{2}/2 & \sqrt{2}/2 \end{vmatrix} = 1 \neq 0$$

Note that since $J = 1 \neq 0$, the following transformation that we're going to consider is OK to go:

(3) $$J = \begin{vmatrix} \sqrt{2}/2 & \sqrt{2}/2 \\ -\sqrt{2}/2 & \sqrt{2}/2 \end{vmatrix} = \begin{vmatrix} \xi_x & \xi_y \\ \eta_x & \eta_y \end{vmatrix} \Rightarrow \left.\begin{matrix} \xi_x = \frac{\sqrt{2}}{2}, & \xi_y = \sqrt{2}/2 \\ \eta_x = -\sqrt{2}/2, & \eta_y = \sqrt{2}/2 \end{matrix}\right\} \Rightarrow$$

(4) $$\begin{cases} \xi(x,y) = \frac{\sqrt{2}}{2}x + \frac{\sqrt{2}}{2}y \\ \eta(x,y) = -\frac{\sqrt{2}}{2}x + \frac{\sqrt{2}}{2}y. \end{cases}$$

Also, by chain rule, we have

(5) $$\begin{cases} u_x = \frac{\partial}{\partial x}\left[u(\xi(x,y), \eta(x,y))\right] = u_\xi \cdot \xi_x + u_\eta \eta_y \quad \& \\ u_{xy} = \frac{\partial}{\partial y}\left[u_x(\xi(x,y), \eta(x,y))\right] = u_{\xi\xi}\xi_x\xi_y + u_{\xi\eta}(\xi_x\eta_y + \xi_y\eta_x) + u_{\eta\eta}\eta_x\eta_y + u_\xi \overset{\nearrow 0}{\xi_{xy}} + u_\eta \overset{\nearrow 0}{\eta_{xy}} \end{cases}$$

So if we plug our result in (5) into eq-n (1):

now by (5)

$$u_{xy} = u_{\xi\xi}\,\xi_x\,\xi_y + u_{\xi\eta}\,(\xi_x\eta_y + \xi_y\eta_x) + u_{\eta\eta}\,\eta_x\,\eta_y = 0$$

$$\Rightarrow \quad u_{\xi\xi}\cdot\left(\frac{\sqrt{2}}{2}\right)\cdot\left(\frac{\sqrt{2}}{2}\right) + u_{\xi\eta}\underbrace{\left\{\left(\frac{\sqrt{2}}{2}\right)\cdot\left(\frac{\sqrt{2}}{2}\right) + \left(\frac{\sqrt{2}}{2}\right)\cdot\left(-\frac{\sqrt{2}}{2}\right)\right\}}_{=0} + u_{\eta\eta}\cdot\left(-\frac{\sqrt{2}}{2}\right)\cdot\left(\frac{\sqrt{2}}{2}\right) = 0$$

$$\Rightarrow \quad \frac{1}{2}u_{\xi\xi} - \frac{1}{2}\cdot u_{\eta\eta} = 0 \quad \Big\}\cdot 2 \text{ Both sides (BS):}$$

$$\boxed{u_{\xi\xi} - u_{\eta\eta} = 0}$$ establishing the desired result. ∎

Prob 4. (2ST, Ch. 5, 7.7)

Describe the regions in the (x,y) plane where the ↓ PDE's are hyperbolic, parabolic, or elliptic:

(a) $2u_{xx} + 4u_{xy} + 3u_{yy} - u = 0$

(b) $y\,u_{xx} - 2u_{xy} + e^x u_{yy} + x^2 u_x - x = 0$

(c) $y\,u_{xx} + 2x\,u_{xy} + x\,u_{yy} + 3u_x - 4y\,u_y + e^x u = 0.$

Sol-n:

(a) given the PDE

(1) $$\overset{a}{2}\,u_{xx} + \overset{b}{4}\,u_{xy} + \overset{c}{3}u_{yy} - u = 0$$

By classification of 2nd order linear PDE's:

(2) $$\begin{cases} \Delta = b^2 - 4ac > 0 & \text{the eq-n is hyperbolic} \\ \Delta = b^2 - 4ac = 0 & \text{the eq-n is parabolic} \\ \Delta = b^2 - 4ac < 0 & \text{... elliptic.} \end{cases}$$

So for (1)

$\Delta = 4^2 - 4\cdot(3)\cdot(2) = 16 - 24 = -8 < 0 \Rightarrow$ hence,

the eq-n in (1) is <u>elliptic</u>. ∎